



Tshwane University  
of Technology

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## MEMORANDUM

Faculty of Science

Department of Pharmaceutical Sciences

|                   |              |
|-------------------|--------------|
| STUDENT NAME      | Lumati Mentz |
| STUDENT NUMBER    | 22 300 3907  |
| STUDENT SIGNATURE |              |

|                                       |                                                    |
|---------------------------------------|----------------------------------------------------|
| DEGREE                                | Bachelor of Health Sciences in Clinical Technology |
| COURSE                                | Pharmacology III in: Reproductive Biology          |
| COURSE CODE                           | RBH317P                                            |
| PAPER                                 | Test 2                                             |
| STANDARD / RE-EXAMINATION             | Standard                                           |
| DATE                                  | 14 April 2025                                      |
| DURATION                              | 80 minutes                                         |
| MARKS                                 | 75                                                 |
| NUMBER OF STUDENTS                    | 1                                                  |
| NUMBER OF PAGES (including this page) | 12                                                 |

|                    |                    |
|--------------------|--------------------|
| INTERNAL EXAMINERS | Mr. K Serabele     |
| MODERATOR          | Mr. William Modiba |

|                              |               |                |
|------------------------------|---------------|----------------|
| SECTIONS                     | MAXIMUM MARKS | MARKS RECEIVED |
| A: Multiple choice questions | 15            |                |
| B: True and False            | 15            |                |
| C: Fill in questions         | 20            |                |
| D: Written Answers           | 25            |                |
| TOTAL                        | 75            |                |

|                                                                                |  |  |
|--------------------------------------------------------------------------------|--|--|
| <b>INSTRUCTIONS TO CANDIDATES</b>                                              |  |  |
| 1. Complete/attempt all the questions. Read all the questions carefully.       |  |  |
| 2. Answer all sections of the examination in the spaces provided on the paper. |  |  |
| 3. Each correct answer/fact will count one mark for section B.                 |  |  |

**TOTAL SECTION A: 15 Marks**

**SECTION A: Multiple Choice Questions**

**IMPORTANT NOTE:**

1. Circle each correct statement
2. Each correct answer will receive plus ONE mark
3. Each omitted answer will receive zero

**A. Drug Therapy in Obstetrics**

**1. Which medication is commonly used for postpartum hemorrhage?**

- a) Ibuprofen
- ☒ b) Oxytocin
- c) Acetaminophen
- d) Labetalol

**2. What is the main concern with using ACE inhibitors during pregnancy?**

- ☒ a) Foetal harm
- b) Maternal hypertension
- c) Preterm labour
- d) Postpartum pain

**3. Which drug is used for seizure prevention in severe preeclampsia?**

- a) Labetalol
- b) Nifedipine
- ☒ c) Magnesium sulphate
- d) Hydralazine

4. Which medication is used for cervical ripening and labor induction?

☒ a) Misoprostol

b) Ibuprofen

c) Acetaminophen

d) Loratadine

5. What is the role of oxytocin in labour induction?

a) To prevent postpartum hemorrhage

b) To enhance milk production

c) To manage pain

☒ d) To stimulate uterine contractions

### B. Special Topics

1. What was involved in the first successful IVF?

a) Unstimulated cycle

b) Laparoscopic retrieval of a single oocyte

c) In vitro fertilization

☒ d) All of the above

2. What is the most common ART procedure?

a) Cryopreservation

b) Intracytoplasmic sperm injection (ICSI)

☒ c) In vitro fertilization (IVF)

d) Intrauterine inseminations

3. Which of the following is excluded from ART?

- a) Procedures manipulating only sperm
- b) Procedures with ovarian stimulation but no egg retrieval
- ☒ c) Both a and b
- d) None of the above

4. What does IVF technology involve?

- a) Unstimulated cycle
- ☒ b) Laparoscopic retrieval of a single oocyte
- c) In vitro fertilization
- d) All of the above

5. What does ART include?

- ☒ a) Fertility treatments where eggs or embryos are manipulated
- b) Procedures manipulating only sperm
- c) Procedures with ovarian stimulation but no egg retrieval
- d) All of the above

### C. Ethical and Societal Considerations

1. What is shared decision-making in healthcare?

- ☒ a) A model of health communication with increasing emphasis
- b) A method of patient isolation
- c) A technique for diagnosing diseases
- d) A way to prescribe medication

**2. What is the primary goal of shared decision-making in healthcare?**

- a) To prescribe medication
- ☒ b) To help the patient make a decision that aligns with their preferences
- c) To diagnose diseases
- d) To isolate patients

**3. What is the importance of dual protection in contraceptive counseling for individuals with HIV/AIDS?**

- a) It prevents pregnancy
- b) It improves health status
- c) It enhances patient engagement
- ☒ d) It reduces the risk of sexually transmitted infections

**4. What is a greatest challenge in contraceptive counseling in South Africa?**

- a) Lack of contraceptive methods
- b) Cultural diversity and varying levels of health literacy
- c) High cost of contraceptives
- ☒ d) Limited access to healthcare facilities

**5. What is the significance of personalized contraceptive counseling in South Africa?**

- ☒ a) It addresses socioeconomic disparities
- b) It prescribes medication
- c) It isolates patients
- d) It diagnoses diseases

# SECTION B: True/ False Questions

TOTAL SECTION B: 15 Marks

## IMPORTANT NOTE:

1. Mark each statement as either **True** or **False** in the box provided to the left of the statement
2. Each correct answer will receive plus ONE mark
3. There will not be negative marking
4. Each omitted answer will receive zero

## Question 1

### Drug Therapy in Obstetrics

|          |                                                        |       |
|----------|--------------------------------------------------------|-------|
| A        | Teratogens pose a risk to the foetus during pregnancy. | False |
| B        | Labetalol is an ACE inhibitor used during pregnancy.   | True  |
| C        | Ice packs are used for uterine cramping management.    | False |
| D        | Breastfeeding provides immune protection for infants.  | True  |
| E        | Acetaminophen is used for cervical ripening.           | False |
| Total: 5 |                                                        |       |

## Question 2

### Special Topics

|          |                                                                                  |       |
|----------|----------------------------------------------------------------------------------|-------|
| A        | DMPA (Depo-Provera) is a short-acting progestin.                                 | False |
| B        | LARC methods include IUDs                                                        | True  |
| C        | "Perfect use" of abstinence is highly effective, but "typical use" is often low. | True  |
| D        | Paediatricians should adopt a "tiered" approach to contraceptive counselling.    | True  |
| E        | DMPA increases estradiol concentrations                                          | False |
| Total: 5 |                                                                                  |       |

## Question 3

### Ethical and Societal Considerations

|          |                                                                                                                            |       |
|----------|----------------------------------------------------------------------------------------------------------------------------|-------|
| T/F      |                                                                                                                            |       |
| A        | Contraceptive counselling in South Africa does not consider individual health status.                                      | False |
| B        | Drug interactions between contraceptives and antiretroviral therapy are a key consideration for individuals with HIV/AIDS. | True  |
| C        | Healthcare providers make decisions for the patient in shared decision-making.                                             | False |
| D        | Building trust and rapport is important in contraceptive counselling.                                                      | True  |
| E        | Shared decision-making is not relevant for preference-sensitive decisions.                                                 | False |
| Total: 5 |                                                                                                                            |       |

Section C: Fill in questions

TOTAL SECTION C: 20 Marks

Fill in Questions

1. Procedures manipulating only sperm are excluded from ART. (1)

2. ART procedures include intra-uterine inseminations. (1)

3. Glucocorticoids are administered in preterm labour to enhance foetal lung maturation. (1)

4. Oxytocin is commonly used for postpartum haemorrhage. (1)

5. Oxytocin stimulates uterine contractions during labour induction. (1)

6. Drug interactions between contraceptives and antiretroviral therapy are a key consideration for individuals with HIV/AIDS. (1)

7. Healthcare providers contribute medical knowledge in shared decision-making. (1)

8. Abstinence is a 100% effective method for preventing pregnancy and STDs when practiced consistently. (2)

9. South Africa's diverse cultural and spiritual beliefs significantly influence perspectives on assisted reproductive technologies and PAR. (2)

10. Advances in reproductive technologies bring new possibilities, leading to complex ethical and legal dilemmas. (2)

11. Advantages of DMPA include protecting against iron-deficiency anaemia and endometrial cancer (2)

12. The estrogen component in the COCs is usually monophasic and biphasic in doses of 10 to 50 µg. (2)

13. Vaginal Ring releases 15 µg ethinyl estradiol and 120 µg etonogestrel daily. (3)

## Question 1

[5]

## 1.1 Explain the pharmacokinetics and pharmacodynamics changes during pregnancy.

The pharmacodynamic process for an individual is straight forward: administration → distribution → metabolism → excretion.

During pregnancy, this process is multiplied and new pharmacokinetics (drug characteristics) are required to adhere to the placenta and fetus.

The placenta is a lipid membrane that acts as a barrier between maternal and fetal circulation. It absorbs drugs through passive diffusion under the following characteristics:

continued on the back of page 9

[5]

## Question 2

## 2.1 Discuss the role of progesterone therapy in preventing preterm labour.

- Low progesterone levels are associated with preterm labour.
- Progesterone inhibits pro-inflammatory cytokines, thus preventing cervical ripening.
- Cervical ripening → the shortening and softening of the cervix.

There are 2 progesterone drug options:

- ~~Hemoxypregesterone~~ caproate  
↳ Prevents preterm labour in women who have experienced it in previous pregnancies.
- Vaginal progesterone  
↳ Prevents cervical shortening.

### Question 3

[5]

3.1 Explain the significance of intracytoplasmic sperm injection (ICSI) in the context of ART.

- ICSI is the most advanced ART procedure, performed by a trained and qualified embryologist.
- The procedure is associated and performed <sup>are present</sup> using when poor quality sperm or oocyte. The quality of these gametes bring down the success rate of other ART procedures (IUI or IVF).
- ICSI provides a sense of hope and better outcome for couples/individuals seeking offspring.
- ICSI uses the highest quality of ART technologies:
  - causing disadvantages such as high cost and great experience of the embryologist
  - But allows for high success rates

### Question 4

[5]

4.1 Discuss the importance of personalized contraceptive counselling in South Africa.

- Contraception is not a "one size fits all" scenario. It is unique (personalized) for each individual:
- Ensures no negative drug interactions between the contraceptive and other chronic medications for HIV, TB, diabetes etc.
- Safe and sensible dosage in prescribed.
- Establishes scheduling routine and follow-up appointments
- Enhances understanding and adherence of the patient towards the medication.
- Enhances provider-patient relationship by building trust
- The provider can empower the decision making process of the patient.

Question 5

[5]

5.1 What are the challenges in contraceptive counselling in South Africa?

- Diverse cultural beliefs
- Diverse languages and communications
- Level of access to ~~the~~ healthcare resources.
- Poverty (high costs of medication / transport)
- Adolescent engagement
- age-appropriate information shared
- various legal restraints regarding age.

TOTAL SECTION A + B + C + D: 75 Marks